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IV CSD-A Major Project Review – Literature Review

A.Y. 2026-2027

Date: 01-10-2026

Team No	A20	
Domain / Areas	Artificial Intelligence/Education	
Title of the Project	CampusIQ: An AI-Powered University Student Support and Academic Intelligence Platform.	
Team Leader Name	Shaik Abdul Kareem Ayemen (23R21A6754)	
Team Members	**Roll No**	**Name**
	23R21A6704	Sai Krishna
	23R21A6755	Shaik Sameer Basha
Guide Name	Arshiya Begum	
Guide Signature		
No. of Papers Reviewed	10	
Period Covered	From 2024 to 2026 (years of publication)	
Sources / Databases Searched	IEEE Xplore, Google Scholar, ResearchGate, J-GLOBAL	
Search Keywords	Retrieval-Augmented Generation (RAG), University AI Assistant, Academic Chatbot, Educational Chatbot, Large Language Models, Semantic Search, University Support System, Student Support, Generative AI in Education, AI-powered	

PROBLEM STATEMENT / SCOPE OF THE SURVEY:

The survey focuses on existing **AI-powered educational assistants, university chatbots, RAG-based question-answering systems, and intelligent student support platforms**. Universities generate large amounts of academic and administrative information through regulations, examination policies, circulars, placement guidelines, and other institutional documents. Existing systems primarily focus on retrieving information or answering student queries, while challenges remain in **personalization, reliable source grounding, intent-based query handling, security, real-time assistance, and integration with student-specific academic information**. The literature survey therefore examines RAG, LLMs, semantic retrieval, hybrid chatbot architectures, personalized academic assistance, and intelligent university support systems to identify gaps that can be addressed by CampusIQ.

LITERATURE SURVEY

S. No.	Title of the Paper, Authors	Journal / Conference, Year	Methodology / Technique Used	Dataset Used	Key Findings / Results	Limitations / Research Gap
1	**Chatbots in Academia: A Retrieval-Augmented Generation Approach** — Maryamah et al.	KST, 2024	RAG, embeddings, LLM-based retrieval	University academic documents	Improved access to academic information through contextual responses.	Mainly focuses on information retrieval; lacks personalized student services.
2	**PUAA: Personal University AI Assistant using Retrieval Augmented Generation** — S. Panday et al.	IEMCON, 2024	RAG, LangChain, LLM	University information documents	Provided quick and contextual answers to university-related queries.	Limited integration with student records and proactive services.
3	**AI-Driven Academic Assistant: An Intelligent Chatbot System Leveraging Llama-3 and RAG** — A. Abdulhussain et al.	CISCT, 2024	Llama-3, RAG, FastAPI, LangChain, Chroma	Multi-format university documents	Improved academic information retrieval and chatbot assistance.	Does not provide complete academic and administrative workflows.
4	**An Intent-Filtered RAG Chatbot for University Admissions in Vietnam** — H. T. Nguyen et al.	ATC, 2024	Intent classification, RAG, semantic retrieval	University admission information	Intent filtering improved retrieval relevance for admission queries.	Restricted mainly to the university admission domain.
5	**A Hybrid Chatbot Model for Enhancing Administrative Support in Education** — K. Mikael et al.	IEEE Access, 2025	Rule-based system, retrieval, generative AI	Educational administrative queries	Hybrid approach improved handling of administrative questions.	Limited personalization and student-specific decision support.
6	**LLM-Based Chatbot for the Indonesia University IT Service Desk** — R. Pratami et al.	ICoDSA, 2025	DeepSeek-v3, RAG, embeddings	University IT support knowledge base	Provided efficient and accurate IT support responses.	Focuses on IT services rather than complete university student support.

S. No.	Title of the Paper, Authors	Journal / Conference, Year	Methodology / Technique Used	Dataset Used	Key Findings / Results	Limitations / Research Gap
7	**AI Chatbots vs. Traditional Search: A Comparative Study on Student Information Retrieval** — L.	FIE, 2025	AI chatbot, RAG, comparative user study	Student information queries	Chatbots improved information understanding and summarization.	Limited study scope and student information domains.
8	**LearnForge: An AI-Powered Intelligent Learning Assistant with RAG-Based Document Intelligence** — N.	ICISCN, 2026	Hybrid retrieval, RAG, reranking	Educational/ learning documents	Improved retrieval and access to learning content.	Primarily focuses on learning materials, not administrative services.
9	**An On-Premise Multilingual Academic Chatbot using RAG and Context-Aware Memory** — M. S. Yaswanth et al.	ICSSIT, 2026	RAG, ChromaDB, memory, multilingual processing	Academic documents	Supported multilingual and context-aware academic assistance.	Limited integration with student portals and university workflows.
10	**SecureContext EDU: An Open Source University Assistant using Dual-LLM Architecture** — G. Kavitha et al.	ICSSCS, 2026	Dual-LLM, RAG, security layer	University knowledge resources	Improved secure and controlled university information assistance.	Strong security focus but limited personalized student assistance.

SUMMARY OF RESEARCH GAPS IDENTIFIED:

Existing studies mainly focus on RAG-based university chatbots and information retrieval. However, limited work integrates institutional knowledge with student-specific academic data, eligibility analysis, personalized recommendations, notifications, grievance management, and administrative workflows.

PROPOSED OBJECTIVES / APPROACH:

CampusIQ addresses these gaps by integrating RAG, NLP, LLMs and semantic search with student-specific data to provide personalized academic and administrative assistance. The system also supports eligibility analysis, recommendations, notifications, grievance management and centralized university information access.

GUIDE REMARKS:

Signature of the Coordinator **Signature of the HOD(CSE-DS)**